SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA0704 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I HARISH KUMAR /192511486(Name / Reg No) certify that thissubmission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary

action. ***Signature of the student:***

Harishkumar

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → → Transport → Network → → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with

their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSIlayers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts;some inaccuracies	Unable to identify key concepts correctly

Concept Mapping Structure	25%	Constructs a clear, logical, and well organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - CONCEPT MAPPING (ANSWER KEY)

Course Name: Computer Networks Course Code: CSA07

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks (BL2, BL3)

Question 1

A partially completed concept map is given: Application → Presentation → → Transport → Network → → Physical. Analyze the missing layers and explain how their functions are essential for reliable communication.

Concept Map

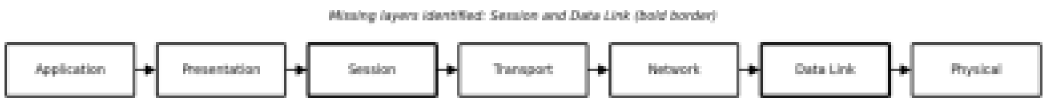

Figure 1: Complete OSI 7-layer concept map with the two missing layers identified

Answer

The two missing layers in the given map are the Session Layer (between Presentation and Transport) and the Data Link Layer (between Network and Physical). The complete chain is:

Application → Presentation → Session → Transport → Network → Data Link → Physical.

- **Session Layer:** Establishes, manages, and terminates the communication session (dialogue) between two end applications. It handles synchronization through checkpoints so that, if a connection is interrupted, communication can resume from the last checkpoint instead of restarting. This directly supports reliability because it prevents complete loss of data during long transfers.
- **Data Link Layer:** Provides node-to-node (hop-by-hop) delivery of frames using physical/MAC addressing, performs error detection through techniques such as CRC, and applies flow control between adjacent devices. It ensures that the raw bit stream from the Physical layer is organized into error-checked frames before being handed to the Network layer.

Together, all seven layers form a functional pipeline: the Application layer generates data, Presentation formats/encrypts it, Session manages the dialogue, Transport ensures end-to-end reliability (sequencing, retransmission), Network provides logical addressing and routing, Data Link ensures reliable hop-to-hop transfer, and Physical transmits raw bits. Reliable communication depends on every layer performing its function correctly and handing a properly encapsulated unit to the layer below/above — the absence of the Session or Data Link layer would break session continuity or introduce undetected transmission errors, making the concept map incomplete without them.

Question 2

Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Concept Map

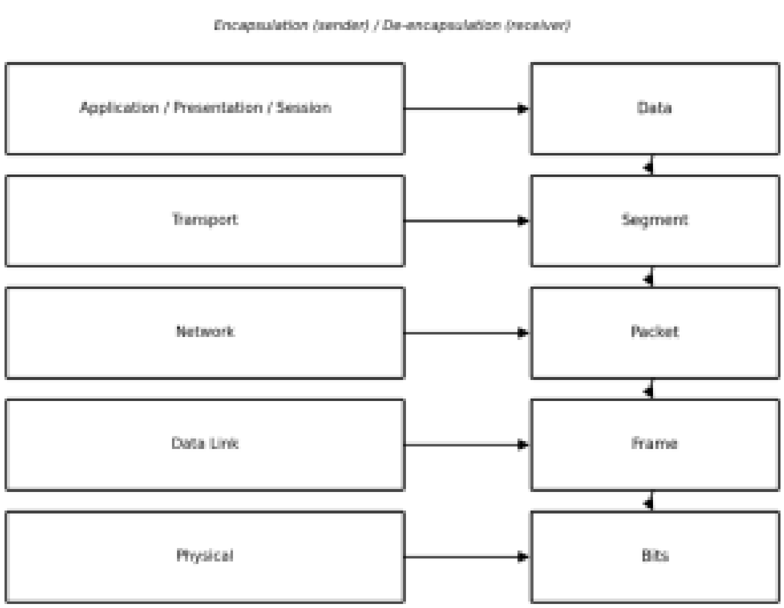

Figure 2: OSI layers mapped to their corresponding Protocol Data Units (PDUs)

Answer

Each OSI layer works with a specific Protocol Data Unit (PDU) as data moves down the stack at the sender (encapsulation) and up the stack at the receiver (de-encapsulation):

- **Application/Presentation/Session Layers – Data:** The user's message is generated and formatted as raw data.
- **Transport Layer – Segment:** Data is broken into segments; a header with source/destination port numbers and sequence numbers is added, enabling reliable, ordered delivery (TCP) or fast delivery (UDP).
- **Network Layer – Packet:** Each segment is encapsulated into a packet with source/destination IP addresses added, enabling logical addressing and routing across networks.
- **Data Link Layer – Frame:** Each packet is encapsulated into a frame with source/destination MAC addresses and a trailer (CRC) added, enabling local delivery and error detection.
- **Physical Layer – Bits:** The frame is converted into a stream of bits (electrical, optical, or radio signals) for transmission over the physical medium.

This transformation of data units is essential for transmission because each layer adds only the control information (header/trailer) it is responsible for, keeping layers independent and modular. At the receiver, the reverse process (de encapsulation) strips each header in the opposite order, allowing the original data to be reconstructed accurately. Without this systematic transformation, addressing, error checking, and routing information could not be attached and interpreted correctly, making end-to-end reliable transmission impossible.

Question 3

Given the concept map: Data Link → MAC Address → Local Delivery; Network → IP Address → Global Delivery. Explain how these relationships ensure complete data delivery across networks.

Concept Map

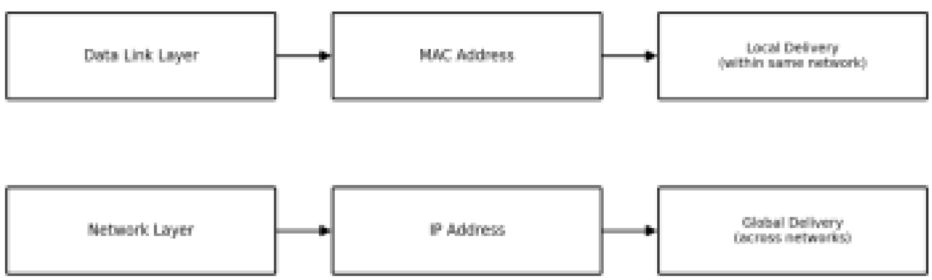

Figure 3: Two-tier addressing model — local delivery vs global delivery

Answer

- **Data Link Layer → MAC Address → Local Delivery:** The Data Link layer uses the physical (MAC) address, a unique 48-bit identifier burned into the network interface card, to deliver frames between devices on the same local network segment (e.g., within a LAN through a switch). MAC addressing is only valid within the local network; it does not extend across routers.
- **Network Layer → IP Address → Global Delivery:** The Network layer uses the logical IP address to identify the source and destination hosts across different networks. Routers use the destination IP address to determine the best path, allowing data to travel across multiple interconnected networks (the Internet) to reach a host that may be on the other side of the world.

These two relationships work together, not independently, to achieve complete delivery. When a packet must cross multiple networks, the IP address (Network layer) remains constant end-to-end and determines the ultimate destination, while the MAC address (Data Link layer) changes at every hop, guiding the frame only to the next intermediate device (switch or router interface) along the path. This is known as the 'next-hop' delivery model: global delivery is achieved through a sequence of local deliveries, each governed by MAC addressing, while overall direction is governed by IP addressing. This layered addressing ensures that data can be delivered efficiently both within a single network and across the wider internetwork.

Question

A concept map shows: Error Control → Data Link Layer → Error Detection; → Transport Layer → Error Correction. Analyze how the interaction between these layers improves reliability in data transmission.

Concept Map

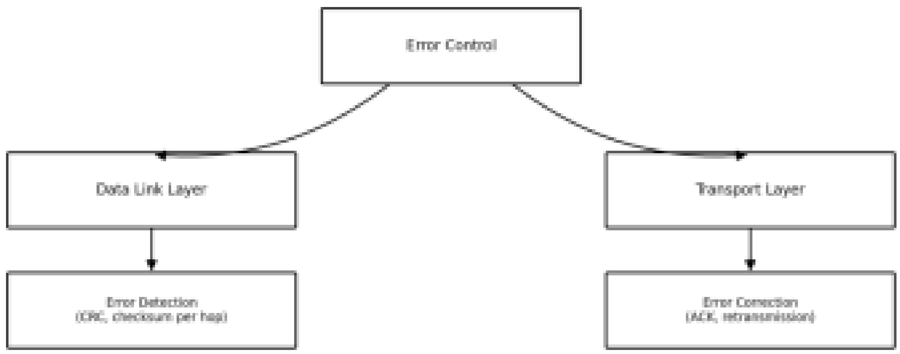

Figure 4: Error control responsibilities distributed across the Data Link and Transport layers

Answer

- **Data Link Layer – Error Detection:** This layer performs error detection on a hop-by-hop basis using techniques such as Cyclic Redundancy Check (CRC), checksums, or parity bits appended to each frame. If corruption is detected during transmission across a single physical link, the frame is discarded, preventing bad data from propagating further into the network.
- **Transport Layer – Error Correction:** This layer provides end-to-end error correction using mechanisms such as sequence numbers, acknowledgements (ACKs), and timers. If a segment is lost, duplicated, or arrives out of order — including cases where the Data Link layer's checks were bypassed or the frame was silently dropped — the Transport layer (e.g., TCP) requests retransmission and reorders segments correctly at the destination.

The interaction between the two layers creates a layered defense: the Data Link layer catches and discards physically corrupted frames close to the point of error (fast, local response), while the Transport layer guarantees that any data lost or corrupted anywhere along the entire end-to-end path — including losses the Data Link layer cannot see, such as congestion drops at intermediate routers — is still recovered through retransmission. Detection alone (Data Link) cannot repair lost data, and correction alone (Transport) would be inefficient if every error had to be resolved end-to-end; together they significantly improve overall reliability while balancing speed and efficiency.

Question 5

Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Concept Map

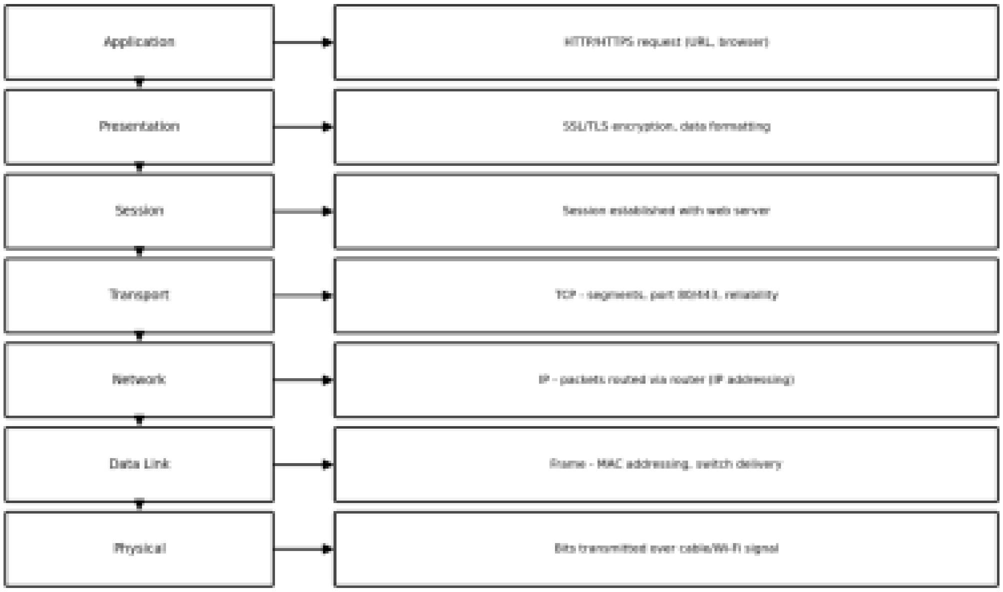

Figure 5: Web browsing session mapped across all seven OSI layers

Answer

When a user browses a website, every OSI layer performs a specific role:

- Application: The browser generates an HTTP/HTTPS request for the requested URL.
- Presentation: Data is encrypted/decrypted using SSL/TLS and formatted (e.g., encoding, compression). ● Session: A session is established and maintained with the web server for the duration of the page load. ● Transport: TCP breaks data into segments, assigns port numbers (80/443), and guarantees reliable, ordered delivery. ● Network: IP addresses the packets and routers determine the path across the internet.
- Data Link: Frames are addressed using MAC addresses for delivery across the local network (e.g., via a switch).
- Physical: Bits are transmitted as electrical, optical, or wireless signals over the medium.

Impact of Layer Failure

Because the layers are sequentially dependent, a failure at any single layer disrupts the entire process:

- **Physical failure** (e.g., cable unplugged, Wi-Fi out of range) — no signal is transmitted at all, so no higher layer can function.
- **Data Link failure** (e.g., MAC conflicts, switch/NIC error) — frames cannot be delivered locally, so the packet never reaches the router even though the physical medium is intact.
- **Network failure** (e.g., misconfigured IP/routing, DNS unresolved) — packets cannot be routed to the correct destination server.
- **Transport failure** (e.g., port blocked, connection reset) — even if packets arrive, no reliable session/data stream can be established with the server.
- **Session/Presentation failure** (e.g., TLS handshake failure) — the browser may show a 'connection not secure' or session error even though lower layers work.
- **Application failure** (e.g., web server down, HTTP error) — the browser receives no valid page content.

This demonstrates that reliable web browsing requires all seven layers to function correctly and pass data upward/downward successfully; the OSI model's layered dependency means a single point of failure anywhere in the chain breaks the complete communication process, which is why systematic, layer-by-layer troubleshooting is essential.

Question 6

A network issue is represented in the concept map: Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖. Analyze the problem and explain how the concept map helps in troubleshooting.

Concept Map

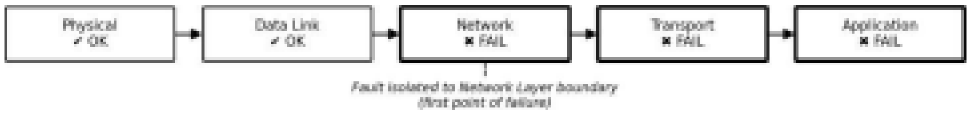

Figure 6: Bottom-up fault isolation showing the point of failure at the Network layer

Answer

The concept map shows that the Physical layer and Data Link layer are functioning correctly (✔), meaning the cable/Wi-Fi signal is present and local frame delivery (MAC-level communication within the LAN, e.g., to the default gateway) is working. However, the Network layer and every layer above it (Transport, Application) are failing (✖).

- **Diagnosis:** Since the fault first appears at the Network layer, the root cause lies specifically at that layer, not below it. Likely causes include an incorrect IP address/subnet mask, a missing or misconfigured default gateway, a routing failure, or a DNS resolution problem. Because the Network layer cannot route packets successfully, the Transport layer cannot establish an end-to-end session, and consequently the Application layer receives no usable data — the failures at Transport and Application are downstream symptoms, not separate root causes.
- **How the concept map helps in troubleshooting:** By representing the OSI layers as a sequential, connected chain, the concept map allows a bottom-up troubleshooting approach: verify each layer starting from Physical and move upward only after confirming the current layer works. Because Physical and Data Link are confirmed working, they can be eliminated from suspicion immediately, saving diagnostic time. The map visually pinpoints the Network layer as the exact boundary where communication breaks down, directing the network administrator to check IP configuration, gateway reachability (e.g., using ping/traceroute), and routing tables rather than wasting effort inspecting cables, NICs, or the application/server itself. This structured, layer-based visualization makes fault isolation faster, more logical, and more accurate than unstructured troubleshooting.